Instructor Guide: Build the OrbitCo KPI Summary

Type: Hands-on lab (individual work) **Time:** 15 minutes (12 min individual work + 3 min share-out)

Pre-Activity Setup

- Confirm every learner successfully completed the Clean the OrbitCo Incident Export lab and has `orbitco-workbook-[yourname].xlsx` open. Learners who are still finishing cleanup should not start this lab — the KPI math depends on the `Tickets` Table being intact.
- Distribute `orbitco_uptime_raw.csv` to every learner via the same channel as the previous raw export.
- Write the three Service Level Agreement (SLA) targets on the board in large text: **High = 60 min, Medium = 240 min, Low = 1440 min.** Learners will need to read these off during step 2 and you will be interrupted three times if they are not visible.
- Have the completed KPI table (from the Example section of the student-facing file) ready to project at the debrief so learners can sanity-check their numbers.

Facilitation Steps

1. **0:00 — Launch** (1 min): Project the Scenario. Name the director's ask out loud ("how many tickets, MTTR overall and by severity, breach count, uptime"). Remind the class: every formula uses a structured reference. If you see a `B2:B500` anywhere, you are about to make a workbook that breaks next week.

2. **0:01 — Circulation, first 4 minutes** (focus: steps 1-5): The `XLOOKUP` step is where the most mistakes happen. Watch for the lookup_array and return_array swapped. If a learner's `SLAMinutes` column returns `#N/A` or "Unknown" on rows that clearly have a severity, check argument order first. The OR-wrapped `SLAStatus` formula also trips people up — they try to nest two `IF`s and miss the empty-string case.

3. **0:05 — Circulation, next 4 minutes** (focus: steps 6-8): The `outputs` sheet is where you find out whether the lesson about structured references landed. Look specifically at the formula bar when you click each KPI cell. If it reads

`=AVERAGEIF(B2:B413, "High", C2:C413)`, make them redo it as `Tickets[Severity]` and `Tickets[ResolutionMinutes]`. The numbers are right; the workbook is wrong.

4. **0:09 — Circulation, last 3 minutes** (focus: steps 9-10): The uptime cell often has the raw decimal showing (0.999) because they forgot the percentage format. The conditional format on `SLAStatus` is the other common miss — if a learner selected the visible `BREACH` cells one by one and applied the rule, the "Applies to" field is a fixed range and the rule will not extend. Catch it by inspecting the rule in Manage Rules.

5. **0:12 — Debrief** (3 min): Project the expected KPI table. Run the three debrief prompts. Close with the forward hook: "This `outputs` sheet becomes the data source for the dashboard you build next lesson."

What to Circulate For

- **Hardcoded ranges instead of structured references.** The single failure mode that matters most. `=COUNTIF(B2:B413, "High")` returns the right number today and the wrong number next Monday. Check every formula bar on the `outputs` sheet.
- **`XLOOKUP` argument order.** The pattern is `XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found])`. A common wrong version is `XLOOKUP([@Severity], SLALookup[SLAMinutes], SLALookup[Severity], "")` — same columns, swapped order. The returned values are numeric-looking (60, 240, 1440) but wrong.
- **Uptime formula hardcoded as `=1 - 43/43200`.** The number is right for this dataset but the formula teaches the wrong habit. Insist on `=1 - (SUM(Uptime[DowntimeMinutes]) / (30*24*60))` even if the answer is identical.
- **Percentage not formatted.** Learners see `0.999` in the uptime cell and think the formula is wrong. It is right; the cell is just displaying a decimal. Fix with Number Format → Percentage → 2 decimals.
- **Conditional format applied to a fixed range.** Open Home → Conditional Formatting → Manage Rules and look at the "Applies to" column. If it reads `=$F$2:$F$414`, redo it by selecting the Table column header first. Correct "Applies to" reads `=Tickets[SLAStatus]`.

- **`Tickets[@Severity]` used outside the Tickets Table.** The `@` row-reference only works inside the same Table. If a learner writes it on the `outputs` sheet, the formula errors. Correct form from another sheet is `Tickets[Severity]` (column reference without `@`).

Common Pitfalls

- **Using `COUNTIFS` with `TRUE` as a text literal.** `=COUNTIFS(Tickets[SeverityIsMissing], "TRUE")` returns 0 because the column contains the boolean `TRUE`, not the string `"TRUE"`. Correct form is `=COUNTIF(Tickets[SeverityIsMissing], TRUE)` — no quotes.
- **`AVERAGEIF` argument order swap.** Pattern: `AVERAGEIF(range_to_test, criterion, range_to_average)`. Very common wrong version: `AVERAGEIF(Tickets[ResolutionMinutes], "High", Tickets[Severity])` — which averages severity text values and returns nonsense or zero.
- **Uptime denominator off by a factor.** Learners divide by `30*24` (getting hours, not minutes) or by `30` (getting days). Only `30*24*60` is correct. Sanity-check: any uptime value outside 99.5–100% on this dataset is wrong.
- **`PROPER` on the severity column eating a second time.** If a learner re-runs the earlier PROPER step on an already-cleaned column, no harm is done; but some learners go back and add PROPER inside `XLOOKUP([@Severity]...)` to "be safe," which then fails when the severity is blank. Leave the clean severity alone.
- **Conditional format rule created with "Use a formula" but the formula references the wrong row.** A common variant: `=[@SLAStatus]="BREACH"`. This works on the current row but only when "Applies to" is the whole Table column. If they used an absolute address like `=$F$2="BREACH"`, every row in the range gets formatted based on row 2's value. Debug by scrolling — if all rows are red or all white, the formula references a single cell.

Debrief Prompts

- **"Add a row of fake data to the `Tickets` Table — a High-severity ticket with ResolutionMinutes = 180. Which outputs updated automatically? Which did not?"** Expected answer: every structured-reference formula updates (total, MTTR overall, MTTR High, High count, SLA breaches — because 180 > 60, this fake ticket

is a breach). Any formula using cell addresses does not. The fake row is the reliability test in miniature.

- **"What does your workbook say about your uptime vs the SLA promise of 99.9%?"** Expected answer: in the 99.85–99.93% range depending on cleanup order — essentially right on the line. This sets up the honest-reporting conversation: OrbitCo is at or just below its SLA promise, and rounding up to 99.9% in a briefing would be optimistic. The dashboard lesson turns this into a traffic-light tile.
- **"If a reviewer asks 'where did the MTTR of 47 come from?', can you trace it back to the raw export?"** Expected answer: yes — `outputs` cell B4 is `=AVERAGE(Tickets[ResolutionMinutes])`, which averages the ResolutionMinutes column on the `Tickets` Table on `tickets_clean`, which derives from a cleaned copy of `tickets_raw`, which was imported from `orbitco_tickets_raw.csv`. Four steps, every one documented. This is the raw-to-processed-to-outputs discipline working as advertised.

Expected Answers

The expected state of the `outputs` sheet after the lab:

KPI (column A)	Approximate Value (column B)	Formula sanity check
Total tickets (30d)	413	Drops below 413 if dedupe was over-eager
Tickets with missing severity	19	Should exactly match the SeverityIsMissing count from the prior lab
MTTR overall (minutes)	~47	Integer or one decimal; not a percentage
MTTR High (minutes)	~38	Lower than overall (High-priority work closes faster)
MTTR Medium (minutes)	~52	Close to overall
MTTR Low (minutes)	~180	Much higher — Low tickets sit longer
High / Medium / Low severity counts	~70 / 210 / 114	Should sum to 413 – 19 = 394 (excluding missing-severity rows)
SLA breaches	In the tens (roughly 30-50)	Never zero; never hundreds
Uptime percentage	99.85–99.93%	Two-decimal percentage format, not a decimal

If a learner's numbers are wildly off (total ticket count ≠ 413, uptime outside the 99.5–100% band, MTTR in the hundreds across all severities), walk them back to the `Tickets` Table and spot-check that the earlier cleanup held — the KPI math assumes a correctly deduplicated, type-converted Table.